Unknown Values, Brackets and Operations

Australian Curriculum v9.0 Year 6 Maths
AC9M6A02

Name: ______________________________ Date: __________________

LEARNING TARGET

I can find unknown values in equations involving brackets and combinations of operations.

KEY STRATEGY + WORKED EXAMPLE

Preserve equality. Simplify known parts using brackets first, then use inverse operations. Substitute the solution into the original equation to verify.

Example: 3 x (n + 4) = 27. Divide by 3: n + 4 = 9. Subtract 4: n = 5. Check: 3 x 9 = 27.

A. TRY TOGETHER - USE THE STRATEGY

1. n + 18 = 45
2. 6n = 54
3. 4 x (n + 2) = 32
4. 72 / (n + 1) = 9

B. CORE PRACTICE - SHOW YOUR THINKING

5. 3n + 5 = 26
6. 5 x (n - 2) = 35
7. 48 / n + 4 = 10
8. 2 x (n + 7) = 30
9. 84 = 7 x (n + 3)
10. (n - 5) / 3 = 4
11. 6 + 4n = 38
12. 100 - 5n = 65

C. INDEPENDENT PRACTICE - CALCULATE AND CHECK

13. 2(n+6)=28
14. 3(n-4)=21
15. 64/n+2=10
16. 90/(n+1)=10
17. 5n-12=33
18. 7+3n=40
19. 4(n+5)-8=32
20. 2[3+n]=18

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D. INDEPENDENT & JUSTIFY

21. A rectangle has perimeter 54 cm and length 16 cm. Write and solve $2(16+w)=54$.

22. Three identical packs plus a $6 fee cost $45. Write and solve an equation for one pack.

23. Correct Isla: From $4(n+3)=28$, she wrote $4n+3=28$.

E. REASON, CHECK AND CORRECT

24. Explain why doing the same operation to both sides preserves equality.

25. Compare solving $5(n+2)=40$ by division first and by expansion. Which is more efficient here?

EXIT TICKET + CHALLENGE

26. Solve $3(2n-1)+4=25$ and verify by substitution.

Challenge: Create an equation involving brackets with solution $n=8$. Exchange with a partner and prove the solution is unique.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

Use your own words so your teacher can see what you understand and what to practise next.

27. Justify the inverse operations and their order.

28. Verify one solution by substituting it into the original equation.

29. Circle one: I understand it | I need more practice | I need help

30. The easiest part was: ____

31. The most difficult part was: ____

Why was it difficult? ____

32. My next practice step is: ____

PARENT / TEACHER TIP

Ask your child to read the equation as a balance. Insist on substitution into the original equation, including brackets, before accepting the answer.

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TEACHER COPY - ANSWERS AND SUPPORT

ANSWER KEY

A. 1) 27. 2) 9. 3) 6. 4) 7.

B. 5) 7. 6) 9. 7) 8. 8) 8. 9) 9. 10) 17. 11) 8. 12) 7.

C. 13) 8. 14) 11. 15) 8. 16) 8. 17) 9. 18) 11. 19) 5. 20) 6.

D. 21) w=11 cm. 22) 3p+6=45; p=$13. 23) 4(n+3)=4n+12, not 4n+3; n=4.

E. 24) Applying the same reversible operation keeps both expressions equal. 25) Divide first: n+2=8, n=6; fewer steps.

EXIT. 26) n=4; check 3(8-1)+4=25.

Challenge: Answers vary. Check that the student's example follows every condition and that the reasoning is mathematically correct.

F. 27-28) Answers vary. Look for a clear strategy and a valid second check. 29-32) Use the reflection to identify confidence, difficulty and the next teaching step.

COMMON MISCONCEPTIONS

Students may ignore brackets, reverse operation order or change only one side. Treat the bracketed expression as one quantity and verify by substitution.

TEACHER CHECK

[ ] Uses an appropriate Year 6 Maths strategy.

[ ] Shows or explains thinking clearly.

[ ] Checks a response using evidence, a rule, or a second method.

[ ] Identifies a specific difficulty and useful next practice step.

PARENT / TEACHER TIP

Ask the child to explain how they know, not only state an answer. If the explanation is unclear, use small objects, a drawing, or a number line before returning to symbols.

NEXT STEP

Revisit the item the child marked difficult. Model one example, solve one together, then ask the child to complete one independently and explain the strategy.